

Balancing User's Friction to Prevent Early Churn

Duolingo Datathon 2026

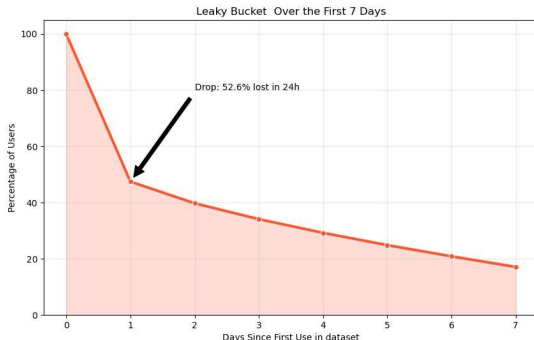


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Problem Statement

A massive segment of the user base quits almost immediately.
In particular, 52.6% of users abandon the app after a single day of use.



Our goal is to understand the reasons that could lead to such a severe drop-off in the beginning.

Friction Definition

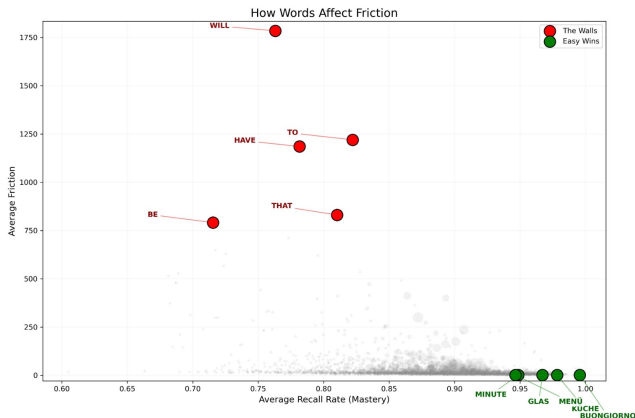
Users are not a monolith; we can split them into two groups: new users and experienced ones (people who use Duolingo repeatedly, who may have built a habit). This is relevant because the two groups respond to the concept of **friction** differently.

We defined **friction** in the following way:

$$friction = \frac{\text{number of times the user has seen a word}}{(\text{observed probability of recalling a word})^2}$$

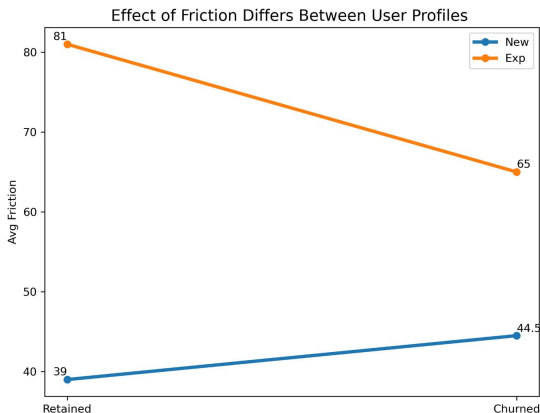
Friction Across Words

To better understand this definition, we look at the level of **friction** across different words, while taking the observed probability of recall into account.



Friction and Churning

The following chart illustrates the **friction** levels across different user profiles:



Friction and Churning

From this plot, we can identify the following:

- *Endowment Effect*:¹ users who invest time and energy integrate the app into their self-identity. For them, the psychological cost of quitting exceeds the "price" of frustration.
- *Loss Aversion*:² people who spent more time in the app built up a streak, which is worth something; they don't want to lose that streak.
- *Learned Industriousness*:³ the "seasoned" users have already experienced the "beauty" friction brings. They do know that "pain leads to gains" and that to functionally learn a language, you have to put in some effort.

¹Kahneman, Knetsch, and Thaler 1991.

²Kahneman, Knetsch, and Thaler 1991.

³Eisenberger 1992.

Possible Causes

The early friction driving the new users away is partly a curriculum design flaw, specifically regarding how different languages introduce vocabulary.

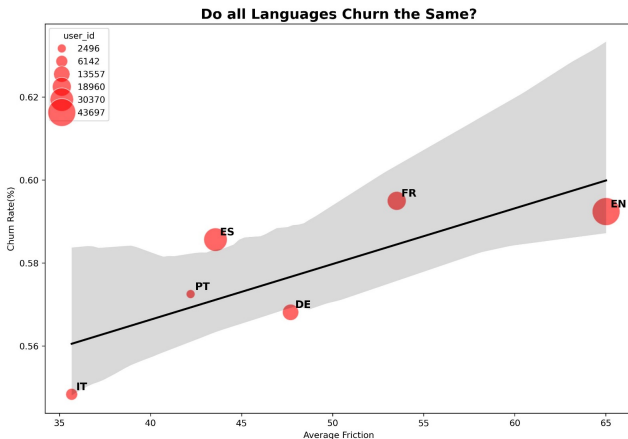
Frontloaded Frustration: Some language tracks introduce complex, difficult words too early in the learning journey.

Noun bias: On the other hand, languages like Italian start with a heavy emphasis on nouns. Nouns are concrete and conceptually easier to grasp. This reduces cognitive load and therefore friction.⁴

⁴Childers 2014.

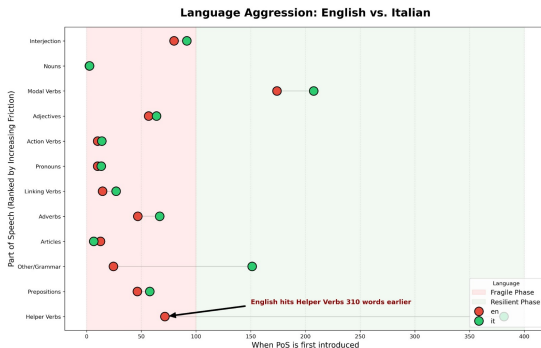
Possible Causes

To identify the possible causes, we first plot the *Churn Rate* across different languages.



Possible Causes

We then compare how early the same type of words are introduced in English and Italian, which displayed divergent patterns in the graph above.



From the plot, we derive that we should strive to engineer the algorithm to reduce friction for new users, as they react worse to increased friction. For that to work, we have to be able to better predict when a new word has to be introduced.

A Data Driven Approach

To find a better solution, we considered **Parts Of Speech (POS)**, which are grammatical categories of words (such as nouns, verbs, adjectives, or adverbs) that differ by their syntactic function and role within a sentence.

This solution does not only come from our graphical analysis, but was also suggested in Settles and Meeder (2016). Unfortunately, we tried to improve the model used in the article by adding this new feature, but the results did not improve. However, we remain confident that this would represent a significant addition to the current curriculum framework.

References I



Childers, Jane B. (June 2014). "Noun Bias in Word Learning". In: *Encyclopedia of Language Development*. Thousand Oaks, CA: SAGE Publications. DOI: 10.4135/9781483346441.n113.



Eisenberger, Robert (1992). "Learned industriousness". In: *Psychological Review* 99.2, pp. 248–267. DOI: 10.1037/0033-295X.99.2.248. URL: <https://doi.org/10.1037/0033-295X.99.2.248>.



Kahneman, Daniel, Jack L. Knetsch, and Richard H. Thaler (Mar. 1991). "Anomalies: The Endowment Effect, Loss Aversion, and Status Quo Bias". In: *Journal of Economic Perspectives* 5.1, pp. 193–206. DOI: 10.1257/jep.5.1.193. URL: <https://www.aeaweb.org/articles?id=10.1257/jep.5.1.193>.

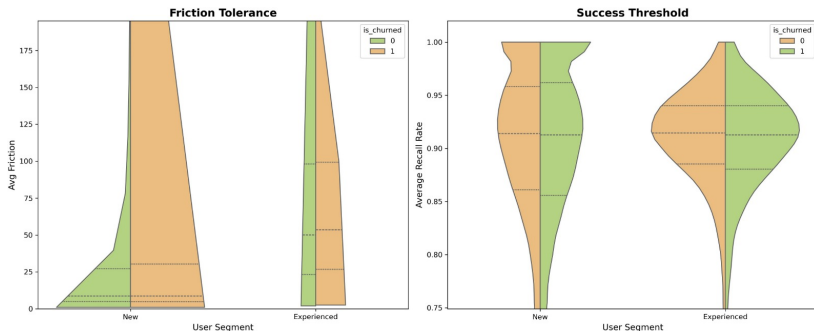


Settles, Burr and Brendan Meeder (Aug. 2016). “A Trainable Spaced Repetition Model for Language Learning”. In: *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Katrin Erk and Noah A. Smith. Berlin, Germany: Association for Computational Linguistics, pp. 1848–1858. DOI: 10.18653/v1/P16-1174. URL: <https://aclanthology.org/P16-1174/>.

Resilience Gap Across Users Profiles

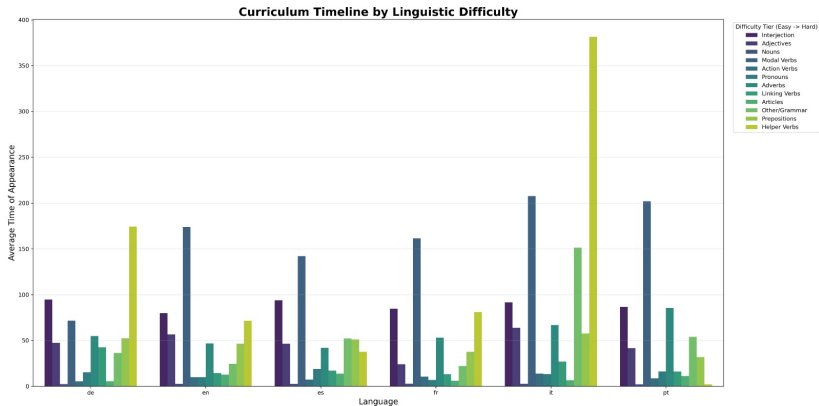
Appendix A

The Resilience Gap: New Vs. Experienced



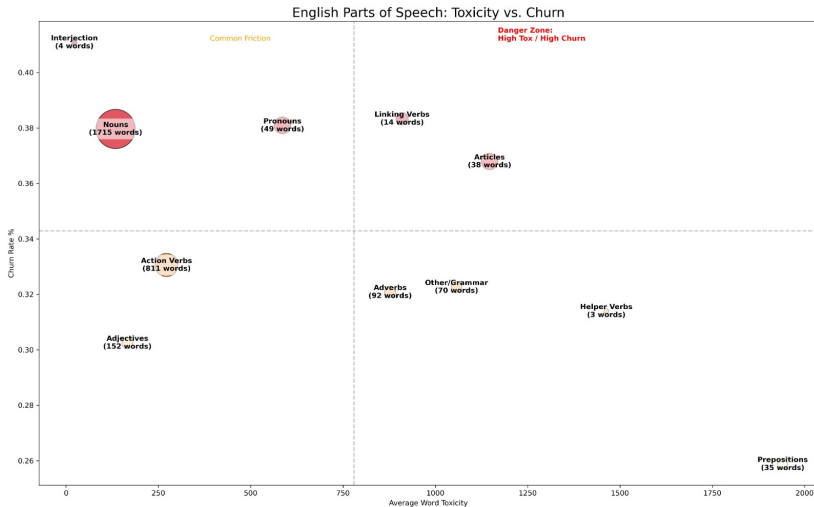
Curriculum Timeline by Linguistic Difficulty

Appendix B



Churn and Toxicity for English Parts Of Speech

Appendix C



Ebbinghaus' Forgetting Curve

Appendix D

